

Crafting the Ultimate Docker Image for Spring Applications



bellsoft

whoami



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BellSoft

- Vendor of Liberica JDK
- Contributor to the OpenJDK
- Author of ARM32 support in JDK
- Own base images
- Own Linux: Alpaquita

Liberica is the JDK officially recommended
by 



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We know our stuff!



So, what is ultimate?

- Smallest — fewest bytes on disk, smallest attack surface
- Fastest startup — 4 seconds or 0.1?
- Something in between — the smallest *pull* per commit

So, let's look at all of
them!

How do we create an image?

Let's start from trivial.

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY . /app
4 RUN cd /app && ./gradlew build
5 ENTRYPOINT java -jar /app/build/libs/spring-petclinic*.jar
```

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Dockerfile directives

1. Each directive creates a layer of the image.
2. Layers are *immutable*
3. Some layers are zero-sized

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1 FROM bash
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Dockerfile directives

1. Each directive creates a layer of the image.
2. Layers are *immutable*
3. Some layers are zero-sized (for example `RUN rm -rf /app`)

```
1 FROM bash
2
3 COPY . /app
4 RUN rm -rf /app
```

4. We would like image to be light, but it's not :(

Result

1	Cmp	Size	Command
2		112 MB	FROM blobs
3		1.9 MB	COPY . /app
4		590 MB	cd /app && ./gradlew build -xtest

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1.9 MB of sources

Result

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590 MB of the app *and* Gradle build caches

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112 MB of Java

1.9 MB of sources

590 MB of the app *and* Gradle build caches

713 MB to ship 1 MB of pet clinic

590 MB are changed on every build!

Why do we care? We care because:

1. `push` takes longer time to start
(update is longer)
2. `pull` takes longer
(update takes longer & scaling takes longer)

Also, more disk space is inefficiently used



Optimizing. Round 1.

Let's build it outside of container

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY . /app
4 RUN cd /app && ./gradlew build
5 ENTRYPOINT java -jar /app/build/libs/spring-petclinic*.jar
```

Optimizing. Round 1.

Let's build it outside of container

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 CMD java -jar /app/app.jar
```

Optimizing. Round 1.

Let's build it outside of container

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 CMD java -jar /app/app.jar
```

Just copying the prebuilt `jar` file

Results

1	Cmp	Size	Command
2		112 MB	FROM blobs
3		1.9 MB	COPY . /app
4		590 MB	cd /app && ./gradlew build -xtest

No Gradle caches and build dir! 713 MB → 187 MB

Results

1	Cmp	Size	Command
2		112 MB	FROM blobs
3		66.4 MB	COPY build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar

No Gradle caches and build dir! 713 MB → 187 MB

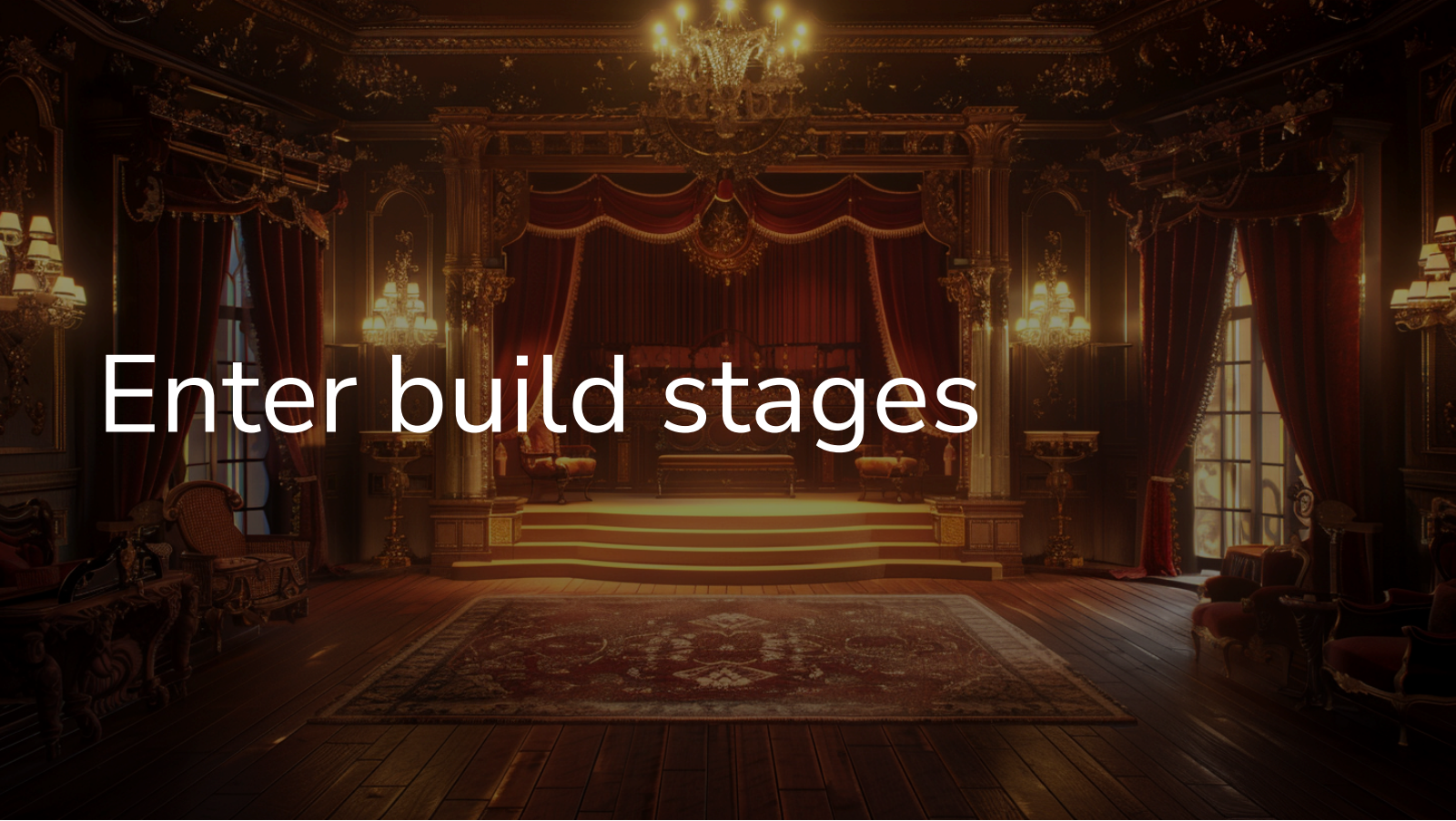
Saved 526 MB

But the build is not clean now :(

Saved 526 MB

But the build is not clean now :(

We have to build everything outside, what if environment affects the build?

A highly detailed, ornate theater stage set. The stage is framed by a grand, classical archway supported by columns. Red velvet curtains hang from the top of the arch. A large, multi-tiered chandelier hangs from the ceiling, casting a warm glow. The stage floor is made of dark wood, and a large, patterned rug is laid out in the foreground. The walls are decorated with intricate carvings and sconces. The overall atmosphere is one of grandeur and elegance.

Enter build stages

Multi-stage builds

Holy grail of pure builds

- Allow clean builds
- Allow optimal packaging
- Allow different base images

Staged example

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-musl as runner
7
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
```

Staged example

```
1  FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
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3  COPY . /app
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6  FROM bellsoft/liberica-runtime-container:jre-26-musl as runner
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8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
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8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
```

Result

- jar — same size, 66.4 MB
- JDK layer 112 MB → JRE layer 144 MB

Result

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Result

- jar — same size, 66.4 MB
- JDK layer 112 MB → JRE layer 144 MB 🤔
- But we *pull* compressed layers: 91 MB → 51 MB
- The JRE has fewer modules (48 vs 68) — it only looks bigger because it ships `lib/modules` uncompressed

That's all folks!

That's all folks!

Or is it?

What's these 66 MB?

We have to pull 66 MB of petclinic on every small commit!

Is there a way to optimize it?

Layers!

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
7
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Diarmode=tools -jar /app/app.jar extract --layers --destination extracted
```

- Build image

Layers!

```
1 FROM bellsoft/liberica-runtime-container:jre-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
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8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
```

- Build image
- Introduce new "optimizer" stage

Layers!

```
4  RUN cd /app && ./gradlew build -x test
5
6  FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
7
8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "app.jar"]
```

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- Introduce new "optimizer" stage

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```
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
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9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "app.jar"]
15 COPY --from=optimizer /app/extracted/dependencies/ /
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure

Layers!

```
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "app.jar"]
15 COPY --from=optimizer /app/extracted/dependencies/ ./
16 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
17 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
```

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Layers!

```
9 WORKDIR /app
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18 COPY --from=optimizer /app/extracted/application/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers!

```
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
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17 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
18 COPY --from=optimizer /app/extracted/application/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
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Layers

Because Spring Boot jar is complex!

```
1  Usage:
2    java -Djarmode=tools -jar my-app.jar
3
4  Available commands:
5    extract      Extract the contents from the jar
6    list-layers  List layers from the jar that can be extracted
```

Layers

```
1  extracted
2  └─ application
3      └─ spring-petclinic-4.0.0-SNAPSHOT.jar
4  └─ dependencies
5      └─ lib
6          └─ angus-activation-2.0.3.jar
7          └─ antlr4-runtime-4.13.2.jar
8          └─ aspectjweaver-1.9.25.1.jar
9          └─ attoparser-2.0.7.RELEASE.jar
10         └─ bootstrap-5.3.8.jar
11         └─ byte-buddy-1.18.10.jar
12         └─ cache-api-1.1.1.jar
13         └─ caffeine-3.2.4.jar
14         └─ classmate-1.7.3.jar
15         └─ commons-logging-1.3.6.jar
16         └─ error_prone_annotations-2.49.0.jar
```


Layers

```
86      — spring-orm-7.0.8.jar
87      — spring-tx-7.0.8.jar
88      — spring-web-7.0.8.jar
89      — spring-webmvc-7.0.8.jar
90      — thymeleaf-3.1.5.RELEASE.jar
91      — thymeleaf-spring6-3.1.5.RELEASE.jar
92      — tomcat-embed-core-11.0.22.jar
93      — tomcat-embed-el-11.0.22.jar
94      — tomcat-embed-websocket-11.0.22.jar
95      — txw2-4.0.9.jar
96      — unescape-1.1.6.RELEASE.jar
97      — webjars-locator-lite-1.1.3.jar
98  — snapshot-dependencies
99  — spring-boot-loader
100
101  6 directories, 93 files
```

Layers

```
86      — spring-orm-7.0.8.jar
87      — spring-tx-7.0.8.jar
88      — spring-web-7.0.8.jar
89      — spring-webmvc-7.0.8.jar
90      — thymeleaf-3.1.5.RELEASE.jar
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Layers

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4  dependencies
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6      └─ angus-activation-2.0.3.jar
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14     └─ classmate-1.7.3.jar
15     └─ commons-logging-1.3.6.jar
16     └─ error_prone_annotations-2.49.0.jar
17     └─ font-awesome-4.7.0.jar
18     └─ h2-2.4.240.jar
19     └─ HdrHistogram-2.2.2.jar
20     └─ hibernate-core-7.4.1.Final.jar
```

Layers

```
1  extracted
2  |— application
3  |   |— spring-petclinic-4.0.0-SNAPSHOT.jar
4  |— dependencies
5  |   |— lib
6  |       |— angus-activation-2.0.3.jar
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```

93 files. This is a *small* Spring app.

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Layered image structure

```
1  > du -h --si --max-depth=1 extracted
2  66M    extracted/dependencies
3  0      extracted/spring-boot-loader
4  0      extracted/snapshot-dependencies
5  1.1M   extracted/application
6  67M    extracted
```

Layered image structure

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2  66M    extracted/dependencies
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5  1.1M   extracted/application
6  67M    extracted
```

- Dependencies: ~66 MB

Layered image structure

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1  > du -h --si --max-depth=1 extracted
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4  0      extracted/snapshot-dependencies
5  1.1M   extracted/application
6  67M    extracted
```

- Dependencies: ~66 MB
- Application: ~1 MB

65.1 MB + 1.1 MB = 66.2
MB



The same 66 MB jar, just cut in two

What are we optimizing?

Pull size!

Every pull after the first one is only ~1 MB — the application layer!

We just reinvented how the BellSoft's buildpack works!

And it is amazing

Digression: buildpacks

<https://paketo.io/>

Trivial usage:

```
1  pack build petclinic \
2    --builder bellsoft/buildpacks.builder:musl \
3    --path build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar \
4    -e BP_JVM_VERSION=26
```

Digression: buildpacks

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2   --builder bellsoft/buildpacks.builder:musl \  
3   --path build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar \  
4   -e BP_JVM_VERSION=26
```

Point it at the ****jar****, not at ``.`` — that is what unlocks the layering

Result

1	ID	TAG	SIZE	COMMAND
2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.59kiB	Buildpacks Launcher Config
4	<missing>		2.79MiB	Buildpacks Application Launcher
5	<missing>		1.38MiB	Application Slice: 4
6	<missing>		392.58kiB	Application Slice: 2
7	<missing>		62.08MiB	Application Slice: 1
8	<missing>		1.06MiB	Software Bill-of-Materials
9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

Result

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2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.59kiB	Buildpacks Launcher Config
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5	<missing>		1.38MiB	Application Slice: 4
6	<missing>		392.58kiB	Application Slice: 2
7	<missing>		62.08MiB	Application Slice: 1
8	<missing>		1.06MiB	Software Bill-of-Materials
9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

The JRE — 111 MB, shared by every app built with this builder

Result

1	ID	TAG	SIZE	COMMAND
2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
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6	<missing>		392.58kiB	Application Slice: 2
7	<missing>		62.08MiB	Application Slice: 1
8	<missing>		1.06MiB	Software Bill-of-Materials
9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

Slice 1 — 65.1 MB of dependencies, untouched between commits

Result

1	ID	TAG	SIZE	COMMAND
2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.59kiB	Buildpacks Launcher Config
4	<missing>		2.79MiB	Buildpacks Application Launcher
5	<missing>		1.38MiB	Application Slice: 4
6	<missing>		392.58kiB	Application Slice: 2
7	<missing>		62.08MiB	Application Slice: 1
8	<missing>		1.06MiB	Software Bill-of-Materials
9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

Slice 4 — 1.45 MB of your classes: the only thing you re-push

Result

1	ID	TAG	SIZE	COMMAND
2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.59kiB	Buildpacks Launcher Config
4	<missing>		2.79MiB	Buildpacks Application Launcher
5	<missing>		1.38MiB	Application Slice: 4
6	<missing>		392.58kiB	Application Slice: 2
7	<missing>		62.08MiB	Application Slice: 1
8	<missing>		1.06MiB	Software Bill-of-Materials
9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

The launcher — 2.93 MB of process types and entrypoint wiring

Result

1	ID	TAG	SIZE	COMMAND
2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.59kiB	Buildpacks Launcher Config
4	<missing>		2.79MiB	Buildpacks Application Launcher
5	<missing>		1.38MiB	Application Slice: 4
6	<missing>		392.58kiB	Application Slice: 2
7	<missing>		62.08MiB	Application Slice: 1
8	<missing>		1.06MiB	Software Bill-of-Materials
9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

226 MB total — and the buildpack sliced it for us

Result

1	ID	TAG	SIZE	COMMAND
2	a3f1c07d21	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.59kiB	Buildpacks Launcher Config
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9	<missing>		105.86MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
10	<missing>		5.10MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack

Our hand-rolled layered image was 214 MB. We paid 12 MB to stop maintaining a Dockerfile.

Buildpacks

Why do we need them?

- Buildpacks transform your application source code into container images
 - The Paketo open source project provides production-ready buildpacks for the most popular languages and frameworks
 - Use Paketo Buildpacks to easily build your apps and keep them updated
- Reminds of s2i images
 - Build better than default
 - Spring-aware

What's still fat?

We made the pull small. What about the thing on disk?

But we promised the
smallest one

So far we only made the *pull* small

Where do those 214 MB live?

1	Size	Layer
2	144 MB	FROM jre-26-slim-musl
3	69 MB	COPY dependencies/
4	1.0 MB	COPY application/

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The JRE is now the biggest thing in the image

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Where do those 214 MB live?

1	Size	Layer
2	144 MB	FROM jre-26-slim-musl
3	69 MB	COPY dependencies/
4	1.0 MB	COPY application/

The JRE is now the biggest thing in the image

Our code is 0.5% of it

Do we need a *whole* JRE?

`jlink` — build a runtime with only the modules you use. Since Java 9.

Ask `jdeps` what the app actually touches:

```
1  jdeps --ignore-missing-deps --print-module-deps -q \  
2    --multi-release 26 -cp 'extracted/dependencies/lib/*' \  
3    --recursive extracted/application/app.jar
```

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2      --multi-release 26 -cp 'extracted/dependencies/lib/*' \  
3      --recursive extracted/application/app.jar
```

```
1  java.base,java.compiler,java.desktop,java.instrument,java.net.http,  
2  java.prefs,java.rmi,java.scripting,java.security.jgss,  
3  java.sql.rowset,jdk.jfr,jdk.management,jdk.net,jdk.unsupported
```


Linking it

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 RUN jlink --add-modules "${jdeps ... app.jar}" \
4       --strip-debug --no-header-files --no-man-pages \
5       --compress zip-9 --output /javaruntime
```

Linking it

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 RUN jlink --add-modules "$(jdeps ... app.jar)" \
4         --strip-debug --no-header-files --no-man-pages \
5         --compress zip-9 --output /javaruntime
```

Linking it

```
1 Error: This JDK does not support linking from the current run-time image
```

The slim JDK ships without `jmods` — `jlink` needs the `-all` one

Linking it

```
1 FROM bellsoft/liberica-runtime-container:jdk-all-26-musl as builder
2
3 RUN jlink --add-modules "$(jdeps ... app.jar)" \
4       --strip-debug --no-header-files --no-man-pages \
5       --compress zip-9 --output /javaruntime
```

The slim JDK ships without `jmods` — `jlink` needs the `-all` one

Linking it

```
1 FROM bellsoft/liberica-runtime-container:jdk-all-26-musl as builder
2
3 RUN jlink --add-modules "${jdeps ... app.jar}" \
4       --strip-debug --no-header-files --no-man-pages \
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```

The slim JDK ships without `jmods` — `jlink` needs the `-all` one

The runtime we got

	Modules	Size
JDK (<code>jdk-all-26-musl</code>)	68	256 MB
JRE	48	144 MB
our <code>jlink</code> runtime	22	77 MB

The runtime we got

	Modules	Size
JDK (<code>jdk-all-26-musl</code>)	68	256 MB
JRE	48	144 MB
our <code>jlink</code> runtime	22	77 MB

Then put it on a base that has nothing else in it

Distroless

```
1 FROM bellsoft/hardened-base:distroless-musl
2 COPY --from=builder /javaruntime /opt/java
3 WORKDIR /app
4 ENTRYPOINT ["/opt/java/bin/java", "-jar", "app.jar"]
5 COPY --from=builder /app/extracted/dependencies/ ./
6 COPY --from=builder /app/extracted/spring-boot-loader/ ./
7 COPY --from=builder /app/extracted/snapshot-dependencies/ ./
8 COPY --from=builder /app/extracted/application/ ./
```


Distroless

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```

0.6 MB of base image. That's the whole OS.

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7 COPY --from=builder /app/extracted/snapshot-dependencies/ ./
8 COPY --from=builder /app/extracted/application/ ./
```

0.6 MB of base image. That's the whole OS.

```
1 > docker run --rm --entrypoint /bin/sh pet-jlinked -c id
2 exec: "/bin/sh": stat /bin/sh: no such file or directory
```

Result

214 MB → 147 MB on disk. 66 MB gone.

Result

214 MB → 147 MB on disk. 66 MB gone.

And it still starts in 2.8 s and still serves pets 🐕

But what about the pull?

But what about the pull?



The pull

Image	On disk	Pulled
layered app, slim JRE	214 MB	111.9 MB
+ distroless base	207 MB	108.8 MB
+ <code>jlink</code> runtime	147 MB	109.7 MB

The pull

Image	On disk	Pulled
layered app, slim JRE	214 MB	111.9 MB
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66 MB off disk — 7 from distroless, 59 from `jlink` — and we pull 2.2 MB less

The pull

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layered app, slim JRE	214 MB	111.9 MB
+ distroless base	207 MB	108.8 MB
+ <code>jlink</code> runtime	147 MB	109.7 MB

66 MB off disk — 7 from distroless, 59 from `jlink` — and we pull 2.2 MB less

The 61.5 MB of dependencies didn't move — `lib/modules` was already compressed

So was it pointless?

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- Pull size: yes, pointless — your dependencies *are* the image

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- Disk and page cache: 66 MB per node, per version

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- Attack surface: 48 modules → 22, no shell, no package manager

So was it pointless?

- Pull size: yes, pointless — your dependencies *are* the image
- Disk and page cache: 66 MB per node, per version
- Attack surface: 48 modules → 22, no shell, no package manager
- And you don't need `jlink` for that last one:

```
bellsoft/hardened-liberica-runtime-container:jre-26-distrolless-  
musl
```

Optimization is multidimensional

When startup time is more important...

There are several solutions for the Spring application startup time

1. Class Data Sharing (CDS)
2. Project Leyden — the AOT cache (JEP 483, JEP 514)
3. Coordinated Restore at Checkpoint
4. Native Image — a whole other talk, and not this one

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

It does not! 🙄

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

It does not! 🤖 But

Application Class Data Sharing

- When: Java 10
- Why: add application classes to the archive

And then:

- When: Java 13, JEP 350
- Why: allow addition of classes into the archive upon app exit!

And this helps! How?

How to use AppCDS with Spring?

- `-XX:ArchiveClassesAtExit=application.jsa` to create an archive
- `-Dspring.context.exit=onRefresh` to start and immediately exit the application

NB:

1. Use the same JDK
2. Use the same arguments

And even better!

Spring AOT

Add `-Dspring.aot.enabled=true`

Even more classes!!!

Practice

```
1 #omitted: builder stage
2 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
3 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 WORKDIR /app
5 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher
6
7 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
8 ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
9 COPY --from=optimizer /app/extracted/dependencies/ ./
10 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
11 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
12 COPY --from=optimizer /app/extracted/application/ ./
```

Practice

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Practice

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3 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 WORKDIR /app
5 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
6
7 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
8 ENTRYPOINT ["java", "-jar", "app.jar"]
9 COPY --from=optimizer /app/extracted/dependencies/ ./
10 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
11 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
12 COPY --from=optimizer /app/extracted/application/ ./
```

Practice

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3 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 WORKDIR /app
5 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
6
7 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
8 ENTRYPOINT ["java", "-jar", "app.jar"]
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11 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
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```

Practice

```
1 #omitted: builder and optimizer stages
2 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
3 ENTRYPOINT ["java", "-jar", "app.jar"]
4 COPY --from=optimizer /app/extracted/dependencies/ ./
5 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
6 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
7 COPY --from=optimizer /app/extracted/application/ ./
8 RUN java -Dspring.aot.enabled=true \
9     -XX:ArchiveClassesAtExit=./application.jsa \
10     -Dspring.context.exit=onRefresh \
11     -jar app.jar
```

Practice

```
1 #omitted: builder and optimizer stages
2 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
3 ENTRYPOINT ["java", "-jar", "app.jar"]
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9     -XX:ArchiveClassesAtExit=./application.jsa \
10     -Dspring.context.exit=onRefresh \
11     -jar app.jar
```

Practice

```
1 #omitted: builder and optimizer stages
2 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
3 ENTRYPOINT ["java", \
4             "-Dspring.aot.enabled=true", \
5             "-XX:SharedArchiveFile=application.jsa", \
6             "-jar", "app.jar"]
7 COPY --from=optimizer /app/extracted/dependencies/ ./
8 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
9 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
10 COPY --from=optimizer /app/extracted/application/ ./
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```

Practice

```
1 #omitted: builder and optimizer stages
2 FROM bellsoft/liberica-runtime-container:jre-25-slim-musl as runner
3 ENTRYPOINT ["java", \
4             "-Dspring.aot.enabled=true", \
5             "-XX:AOTCache=app.aot", \
6             "-jar", "app.jar"]
7 COPY --from=optimizer /app/extracted/dependencies/ ./
8 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
9 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
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14     -jar app.jar
```

What does it cost ?

```
1  > ls -lh --si app.aot  
2  117M Aug 11 13:47 app.aot
```

Which is not small at all!

What does it cost ?

```
1 > ls -lh --si app.aot
2 117M Aug 11 13:47 app.aot
```

Which is not small at all!

But you're trading pull speed for startup speed

Startup, measured

Same host, same layered image — AOT rows from an earlier idle-machine run

Image	JDK	Startup
plain JRE, no cache	26	3.21 – 3.52 s
<code>-XX:SharedArchiveFile</code> (AppCDS)	26	1.80 – 1.99 s
<code>-XX:AOTCache</code>	25	1.12 – 1.18 s
<code>-XX:AOTCache</code> + <code>-Dspring.aot.enabled=true</code>	25	0.69 – 0.79 s

AOTCache throws `InternalError` on every 26 start — our own image. AOT rows stay 25.

Pushing further with CRaC

CRaC: Coordinated Restore at Checkpoint

...checkpoint (snapshot) a Java instance while it is executing. Restoring from the image could be a solution to some of the problems with the start-up and warm-up times.

<https://openjdk.org/projects/crac/>

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CRaC: Coordinated Restore at Checkpoint

...checkpoint (snapshot) a Java instance while it is executing. Restoring from the image could be a solution to some of the problems with the start-up and warm-up times.

<https://openjdk.org/projects/crac/>

- `jre-crac-slim` is JDK 21 — no CRaC build of 26 yet, so this section drops five versions
- the app needs `org.crac:crac` on the classpath for `spring.context.checkpoint` to fire

In a perfect world it should be:

```
1  FROM bellsoft/liberica-runtime-container:jdk-25-musl as builder
2
3  COPY . /app
4  RUN cd /app && ./gradlew build
5
6  FROM bellsoft/liberica-runtime-container:jre-crac-slim as optimizer
7
8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
11
12 FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner
13
14 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
15 COPY --from=optimizer /app/app.jar /app/app.jar
16 COPY --from=optimizer /checkpoint /checkpoint
```


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16 COPY --from=optimizer /checkpoint /checkpoint
```

In a perfect world it should be:

```
1  FROM bellsoft/liberica-runtime-container:jdk-25-musl as builder
2
3  COPY . /app
4  RUN cd /app && ./gradlew build
5
6  FROM bellsoft/liberica-runtime-container:jre-crac-slim as optimizer
7
8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
11
12 FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner
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14 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
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```

But in reality

```
1  #12 6.051      Suppressed: java.lang.RuntimeException: Native checkpoint failed.
2  #12 6.051      at java.base/jdk.crac.Core.translateJVMExceptions(Core.java:114) ~[r
3  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore1(Core.java:192) ~[na:na
4  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore(Core.java:299) ~[na:na
5  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore(Core.java:278) ~[na:na
6  #12 6.051      at java.base/javax.crac.Core.checkpointRestore(Core.java:73) ~[na:na
7  #12 6.051      at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke
8  #12 6.051      at java.base/java.lang.reflect.Method.invoke(Method.java:580) ~[na:r
9  #12 6.051      at org.crac.Core$Compat.checkpointRestore(Core.java:141) ~[crac-1.4.
10 #12 6.051      ... 17 common frames omitted
```

CRaC is hard!

Let's try to fix it with arcane magic

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The checkpoint *works*. It's the exit code that lies — so we tell Docker to ignore it.

The magic needs a magic wand

`docker build` can't grant `security.insecure`. `buildx` can.

```
1 docker buildx create --buildkitd-flags \  
2   '--allow-insecure-entitlement security.insecure' \  
3   --name insecure-builder
```

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```
1 docker buildx use insecure-builder
```

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```
1 docker buildx build \  
2   --allow security.insecure \  
3   -f Dockerfile.crac \  
4   -t pet-crac --output type=docker .
```

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```
1 docker run --rm -it --privileged pet-crack
```

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- 2 HikariPool-1 - Thread starvation or clock leap detected (housekeeper delta=1h6m48s576ms).
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Insecure builder, privileged container. Do not show this slide to your security team.

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Insecure builder, privileged container. Do not show this slide to your security team.

Hikari is right: from the pool's point of view the JVM just skipped an hour.

What does *this* cost?

```
1  443 MB  /checkpoint
2   66 MB  app.jar
3  199 MB  JDK 21 CRaC JRE
4  _____
5  708 MB  total
```

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```

0.1 s startup — 5 MB *under* the naive image we opened with

Is it a good solution?

It depends

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- If "It Works" is enough for you

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- If "It Works" is enough for you -> YES
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 1. `CAP_SYS_PTRACE`
 2. `CAP_CHECKPOINT_RESTORE`
4. Container will run and stop
5. Commit the container like

```
1  docker commit container-id new-tag
```

Pros and Cons

Pros:

1. Does not require arcane magic
2. Works more predictably
3. Does not depend on unstable features
4. Does not require privileged containers in production

Cons:

1. Organization-specific
2. Requires more steps

Four flavours of ultimate, and one we skipped

Quick summary?

1. Use layers for faster deployment — 66 MB pulled once, ~1 MB per commit after
2. Use `jlink` + distroless for the smallest, quietest image — 214 MB → 147 MB, 22 modules, no shell
3. Use CDS or the AOT cache for faster startup without many compromises — 3.5 s → 0.7 s
4. Use CRaC for a *lightning-fast* startup — 0.1 s, for a 708 MB image and privileged builds
5. Native image is the fifth option — ask me at the booth, it needs its own hour

So what is the ultimate image?

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The one you didn't have to think about

So what is the ultimate image?

The one you didn't have to think about
Layers you get for free. Startup you pay for once. Size you pay
for forever.

Thank you!

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END